AI Exposure and Labor Market Displacement Across U.S. Occupations

A Cross-Sectional and Difference-in-Differences Analysis of 673 Occupations, 2019-2024

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Abstract

How does artificial intelligence reshape labor markets when its capabilities target cognitive rather than manual tasks? This paper exploits the Felten, Raj, and Seamans (2021) AI Occupational Exposure Index (AIOE), mapped onto the Bureau of Labor Statistics Occupational Employment and Wage Statistics (May 2019 and May 2024) and O*NET 30.2 task descriptors, to analyze 673 U.S. occupations covering more than 154 million workers. Three findings emerge. First, AI exposure is concentrated at the top of the occupational ladder: the correlation between AIOE and log annual wages is 0.54 ($p < 0.001$), and AIOE remains a significant wage predictor after controlling for education, occupation group, and employment size ($R^2 = 0.64$). Second, exposure is distributionally skewed, female-dominated occupation groups display a mean AIOE of 0.54 versus −0.71 in male-dominated groups ($t = -16.05$, $p < 0.001$). Third, a panel difference-in-differences design exploiting 661 occupations matched across 2019 and 2024 reveals that high-AIOE occupations experienced faster employment growth ($\beta = 0.052$, $p < 0.001$) but slower wage growth ($\beta = -0.029$, $p < 0.001$) than low-AIOE occupations. This pattern is consistent with AI functioning as a labor-augmenting technology that expands the effective supply of cognitive work and compresses skill premiums rather than producing wholesale displacement. Within high-exposure occupations, outcomes are sharply heterogeneous: routine cognitive roles such as credit authorizers (−55%) and telemarketers (−51%) contracted, while managerial and research roles grew (industrial-organizational psychologists +67%, epidemiologists +55%). These findings add a descriptive quasi-experimental data point to an empirical literature that, as Kolko (2026) characterizes it, remains in its earliest innings.

Keywords: artificial intelligence, automation, labor markets, difference-in-differences, occupational exposure, wage dynamics, skill premium compression.

1. Introduction and Motivation

The rapid advancement of artificial intelligence has generated widespread concern about its potential to reshape labor markets. Unlike previous waves of automation, which primarily displaced routine manual and cognitive tasks in middle-skill occupations, modern AI systems demonstrate capabilities that extend into the analytical, creative, and interpersonal domains traditionally considered the province of highly educated workers. Understanding which occupations, industries, and demographic groups face the greatest exposure to AI is therefore a question of both scholarly and policy importance.

This paper asks three related questions. First, how is AI exposure distributed across U.S. occupations, and how does that distribution relate to wages, education, and occupation structure? Second, do female-dominated occupation groups face systematically different exposure than male-dominated ones, and if so, what are the distributional implications? Third, and most importantly, have high-exposure occupations already begun to diverge from low-exposure occupations in employment and wage dynamics during the 2019-2024 window spanning the public release of ChatGPT (November 2022) and GPT-4 (March 2023)?

This project was motivated by research questions that emerged during my ECO483 coursework in Winter 2026, where I studied the relationship between Income Inequality, Social Capital, and Mental Health Distress in U.S. Counties. That work sharpened my interest in how structural economic forces shape labor market outcomes and downstream well-being. The intellectual progression from studying the health consequences of economic inequality to investigating the labor market consequences of technological disruption felt natural: both involve identifying which populations are most vulnerable to large-scale structural change, and both require careful attention to identification challenges and confounding.

The broader research landscape provided additional impetus. The Federal Reserve flagged AI-driven hiring caution in its January 2026 FOMC minutes. In March 2026, Jed Kolko published a comprehensive review at PIIE and the Hamilton Project characterizing research on AI and labor markets as being in its earliest stages, with conflicting evidence on whether high-exposure occupations are already experiencing employment declines. Kolko identified three reasons why current findings are inconclusive: early research on labor demand is collectively mixed, any current findings are weak signals about the future given AI's rapid evolution, and the existing literature focuses narrowly on labor demand while neglecting productivity, labor supply, and transition dynamics. This project contributes to this emerging literature by providing an accessible, replicable analysis that bridges the AIOE framework with current BLS employment data and offers an explicit panel difference-in-differences counterpart to the prevailing cross-sectional evidence.

2. Literature Review

2.1 Theoretical Frameworks: From Routine-Biased to Capability-Aligned Automation

The economic analysis of AI and labor markets builds on the broader literature on automation and technological change. Acemoglu and Restrepo (2020) established that industrial robots reduced employment and wages in exposed U.S. commuting zones. Their task-based framework, which models occupations as bundles of tasks with varying susceptibility to automation, has become foundational. The key insight is that technologies displace workers by substituting for specific tasks, while the employment and wage implications depend on whether displacement is offset by productivity gains, reinstatement of labor into new tasks, or capital-labor complementarities.

Felten, Raj, and Seamans (2021) developed the AI Occupational Exposure Index (AIOE) used in this analysis. The AIOE measures the overlap between the abilities required by each occupation (as defined in O*NET) and the capabilities demonstrated by AI applications across ten domains. A central result from their work is that AI exposure follows a different occupational gradient than earlier automation: highly exposed occupations tend to be cognitive and analytical rather than routine and manual.

2.2 Emerging Empirical Evidence on AI and Employment

Brynjolfsson and Mitchell (2017) found that suitability for machine learning is distributed across the occupational spectrum rather than concentrated at the bottom. Webb (2020) constructed an alternative exposure measure based on patent text overlap, finding that AI-exposed occupations are disproportionately high-skill and high-wage.

The most recent literature, reviewed by Kolko (2026) at PIIE, reveals conflicting evidence. Brynjolfsson, Chandar, and Chen (2025) found declining employment among young workers in AI-exposed occupations using proprietary ADP payroll data, but Eckhardt and Goldschlag (2025) found the opposite using publicly available CPS data. Iscenko and Millet (2026) found that job posting declines in AI-exposed occupations began in 2022, before ChatGPT's release, suggesting macroeconomic factors rather than AI as the driver. Frank et al. (2026) reached similar conclusions. Massenkoff and McCrory (2026) introduced a new measure combining theoretical capability with real-world usage data, finding no systematic increase in unemployment for highly exposed workers but suggestive evidence of slowed hiring for younger workers.

This paper positions itself within this emerging debate by offering a transparent, fully replicable DiD design at the occupation level using public BLS microdata, which complements (rather than duplicates) the young-worker-focused payroll analyses cited above.

3. Data and Methodology

3.1 Data Sources

The analysis draws on three primary datasets obtained directly from their official sources. First, the Felten et al. (2021) AI Occupational Exposure Index (AIOE_DataAppendix.xlsx from GitHub) provides AIOE scores for 774 occupations indexed by six-digit SOC codes. Second, the BLS Occupational Employment and Wage Statistics (national_M2024_dl.xlsx and national_M2019_dl.xlsx from bls.gov/oes/tables.htm) provide employment counts and wage distributions for detailed occupations from the May 2024 and May 2019 surveys. Third, the O*NET 30.2 Database (individual Excel files from onetcenter.org) provides occupation-level descriptors, including education requirements and job zone classifications.

3.2 Harmonization and Sample Construction

Merging across these datasets required harmonizing SOC codes at the six-digit level. After inner-joining AIOE with May 2024 OEWS and dropping observations with missing employment or wage data, the cross-sectional analytical sample contains 673 occupations representing over 154 million workers, of which 659 also carry O*NET education data for the regression analysis. For the panel analysis, I additionally require non-missing employment and mean wage in May 2019 at the six-digit SOC level, yielding a balanced panel of 661 occupations observed in both years.

3.3 Cross-Sectional Regression Specification

Following a progressive model-building strategy, I estimate three OLS regressions of log annual mean wages on AIOE. Model 1 is bivariate. Model 2 adds log employment, dummy variables for required education level (with bachelor's as the reference), and major occupation group fixed effects. Model 3 additionally introduces an AIOE × graduate-education interaction to test whether the AIOE-wage gradient

attenuates at the top of the education distribution. All cross-sectional models are estimated with heteroskedasticity-robust (HC1) standard errors.

3.4 Difference-in-Differences Specification

The panel DiD design stacks 2019 and 2024 observations in long format and estimates, for each outcome $Y \in$ {log employment, log wage}:

$$Y_{it} = \alpha + \beta_1 \cdot AIOE_i + \beta_2 \cdot Post_t + \beta_3 \cdot (AIOE_i \times Post_t) + \varepsilon_{it}$$

where $AIOE_i$ is continuous, $Post_t$ is an indicator for 2024, and β_3 is the DiD parameter of interest, the differential change in the outcome associated with a one-unit increase in AIOE. Standard errors are clustered at the occupation level to account for serial correlation within each SOC code across the two periods. I also report first-difference regressions of $\Delta\log(Y)$ on AIOE, with and without baseline controls for 2019 log employment and log wages, which are numerically equivalent in slope to the DiD interaction.

Identification relies on the standard DiD assumption of parallel trends, that, absent AI deployment, occupations with higher AIOE would have evolved proportionally to those with lower AIOE. With only two periods, this assumption is not directly testable, and I return to this limitation in Section 6. The specification nonetheless represents a meaningful advance over pure cross-sectional analysis because it differentiates out time-invariant occupation-level confounders such as underlying skill composition, task routineness, and career structure.

4. Cross-Sectional Results

4.1 Distribution of AI Exposure

AI exposure varies substantially across the 673 occupations in the sample. The AIOE ranges from −2.67 (Dancers) to 1.53 (Genetic Counselors), with a mean of −0.02 and a standard deviation of 1.01. The most AI-exposed occupations are concentrated in analytical and cognitive domains: genetic counselors, financial examiners, actuaries, budget analysts, judges, and accountants all rank in the top ten. Accountants and auditors (AIOE = 1.48) represent 1.45 million workers, making them the largest high-exposure occupation by employment.

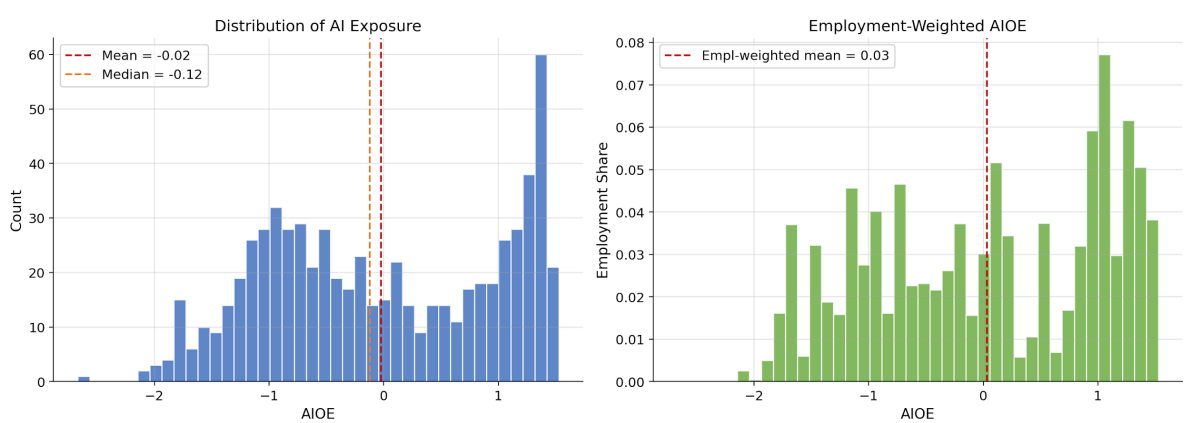

Figure 1. Distribution of AI occupational exposure (AIOE) across 673 U.S. occupations.

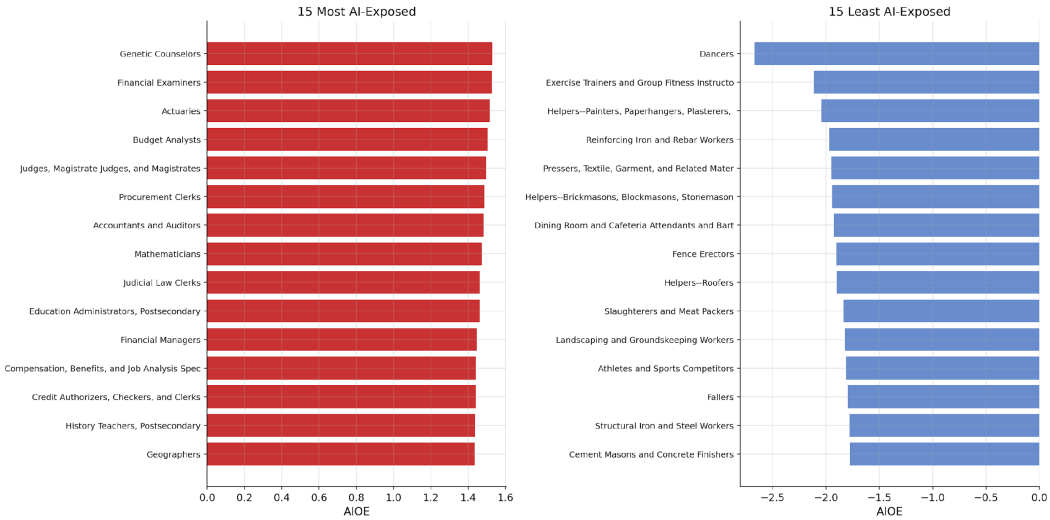

Figure 2. The 15 most and least AI-exposed occupations.

4.2 AI Exposure by Occupation Group

Computer and mathematical occupations have the highest mean AIOE (1.44), followed by legal (1.38), business and financial (1.19), and education (1.08). Construction and extraction (−1.33), cleaning and maintenance (−1.19), and installation, maintenance, and repair (−0.89) are the least exposed. The gradient traces a stark reversal of earlier automation patterns: the occupations most insulated from AI are precisely those whose physical embodiment requirements most resist digital automation.

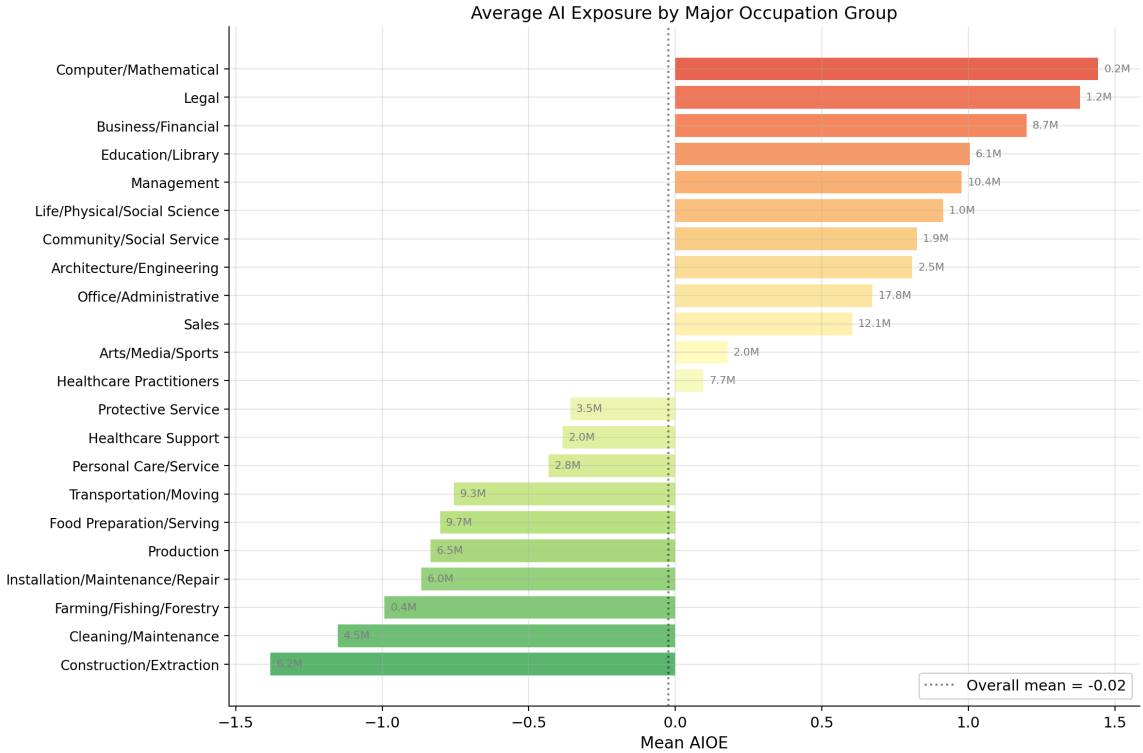

Figure 3. Mean AIOE by major occupation group, with total employment labeled.

4.3 AI Exposure and Wages

There is a strong positive relationship between AI exposure and wages. The Pearson correlation between AIOE and annual mean wages is $r = 0.46$ ($p < 0.001$), and with log wages $r = 0.54$ ($p < 0.001$). The binned scatter plot in Figure 4 shows a clear upward gradient.

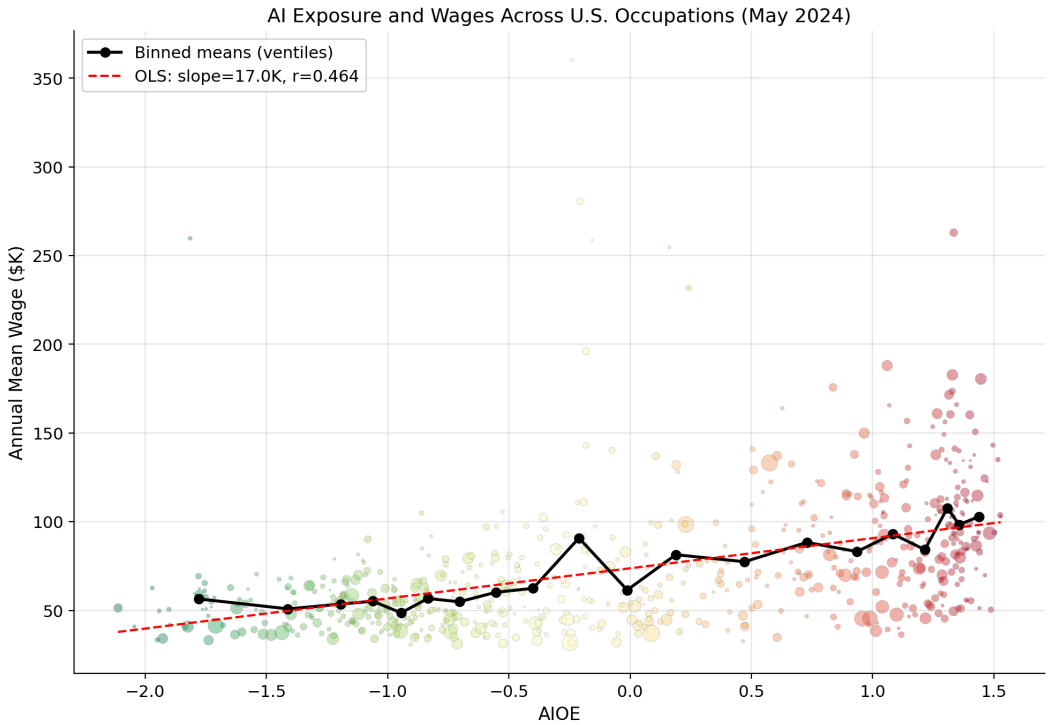

Figure 4. Bivariate relationship between AI exposure and annual mean wages. Bubble size reflects employment.

4.4 OLS Regression Results

Table 1 reports the three progressive specifications. In Model 1, the bivariate regression yields an AIOE coefficient of 0.226 ($p < 0.001$), indicating that a one-unit increase in AIOE is associated with approximately 25% higher wages. AIOE alone explains 29.4% of cross-occupational wage variation.

Model 2 adds controls. The AIOE coefficient attenuates to 0.105 ($p < 0.001$) but remains highly significant, while R^2 increases to 0.640. Occupations requiring only a high school diploma pay 32% less than those requiring a bachelor's degree, while those requiring graduate education pay 23% more. The attenuation suggests that roughly half of the raw AIOE-wage association is mediated by education and occupation structure, but a substantial direct premium persists.

Model 3 introduces the AIOE × graduate-education interaction. The interaction term is negative and significant ($\beta = -0.151$, $p = 0.028$), suggesting the wage premium from AI exposure is weaker for graduate-level occupations, which already command high wages through conventional human capital returns. This diminishing-marginal-return pattern is consistent with AIOE capturing a complementarity effect that saturates at the top of the education distribution.

Table 1. OLS Regression Results: Log(Annual Wage) on AI Exposure

Variable	Model 1	Model 2	Model 3

AIOE	0.226***	0.105***	0.135***
Log(Employment)		−0.016*	−0.016*
HS or Less (ref: Bachelor's)		−0.323***	−0.295***
Some College		−0.162***	−0.123*
Graduate+		0.232***	0.371***
AIOE × Graduate			−0.151*
Occupation Group FE	No	Yes	Yes
R-squared	**0.294**	**0.640**	**0.643**
N	659	659	659

Robust (HC1) standard errors. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

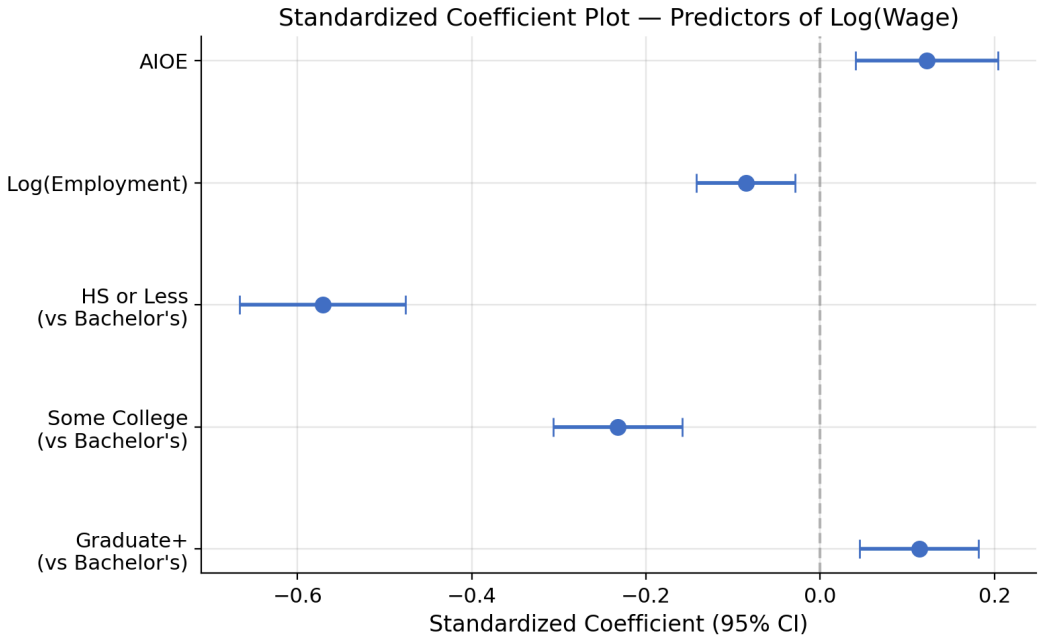

Figure 5. Standardized coefficient plot from Model 2 with 95% confidence intervals.

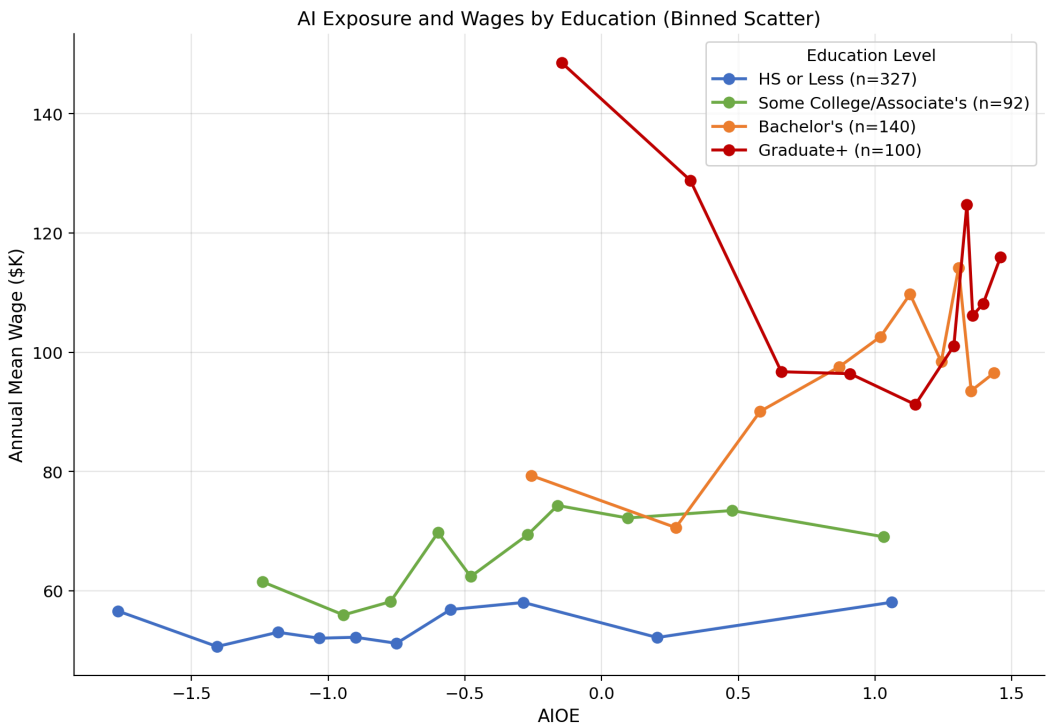

Figure 6. Binned scatter: AIOE and wages by education requirement.

4.5 K-Means Clustering

To move beyond parametric regression and identify natural risk tiers, I apply K-means clustering on a standardized feature vector comprising AIOE, log annual wage, and log employment. Silhouette analysis supports k = 4. The resulting tiers are: Low Risk (219 occupations, mean AIOE = −0.82), Moderate Risk (150 occupations, AIOE = −0.72), Elevated Risk (144 occupations, AIOE = 0.89), and High Risk (157 occupations, AIOE = 0.93). The High-Risk tier, which combines high exposure with very large employment shares, includes general and operations managers, registered nurses, office clerks, and accountants and auditors, representing 54.5 million workers in aggregate.

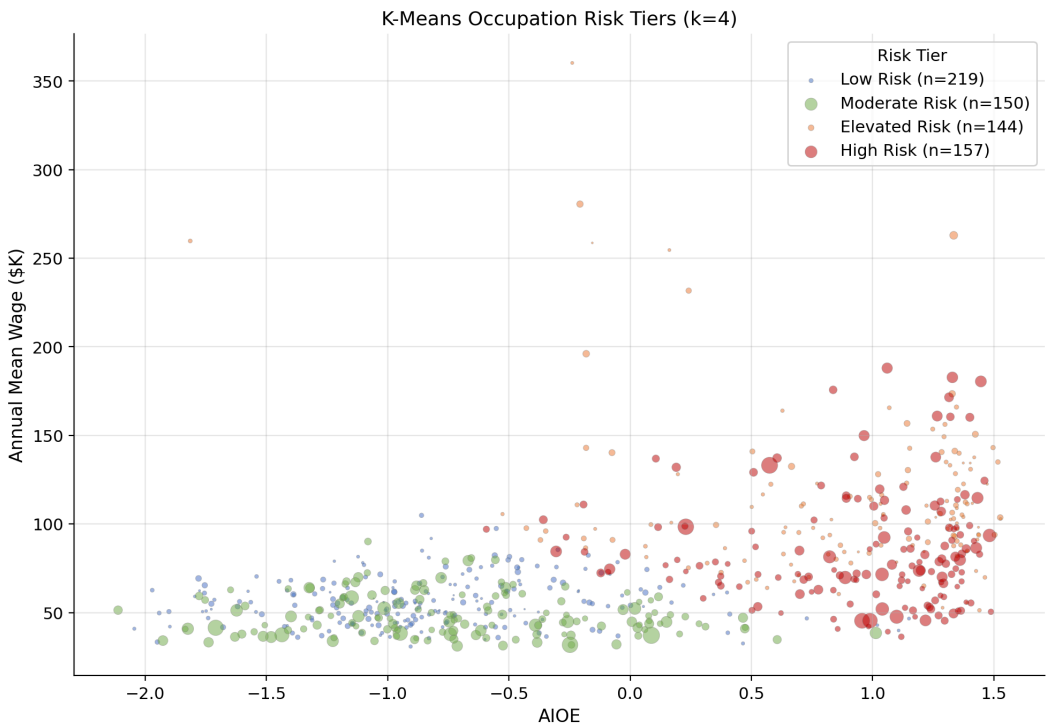

Figure 7. K-means occupation risk tiers (k = 4) projected on AIOE and log annual wage.

4.6 Demographic Vulnerability

Female-dominated occupation groups exhibit significantly higher mean AI exposure (AIOE = 0.54) than male-dominated groups (AIOE = −0.71), with a two-sample t-test confirming the difference ($t = -16.05$, $p < 0.001$). Employment-weighted exposure sharpens this contrast: female-dominated occupations carry a weighted AIOE of 0.65 versus −0.86 in male-dominated groups. This finding raises a distributional concern that is underemphasized in the broader AI-labor discourse: the occupations most exposed to AI disruption, administrative support, healthcare support, and education-adjacent roles, are disproportionately staffed by women, whose historical employment gains since the 1980s have concentrated precisely in these cognitive-service sectors.

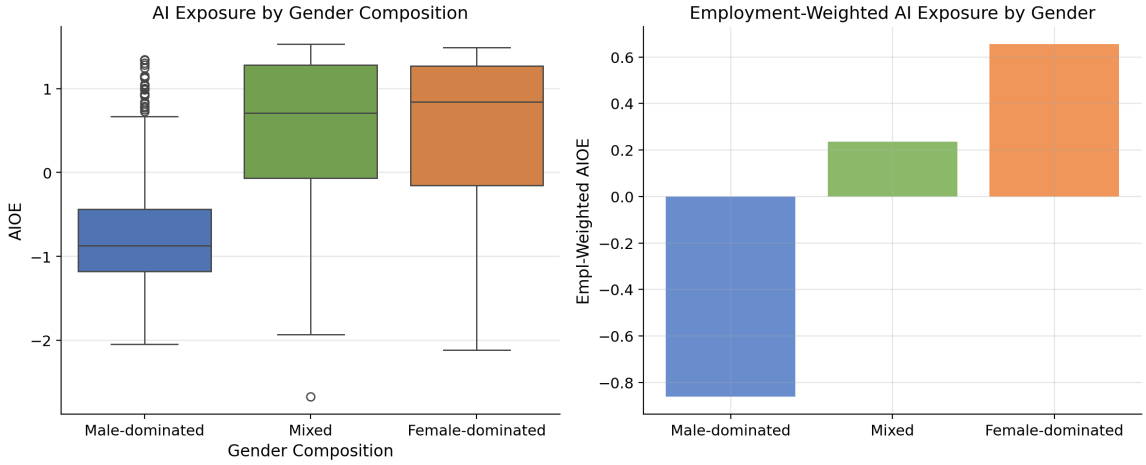

Figure 8. AI exposure by gender composition of occupation groups.

5. Difference-in-Differences Analysis: Employment and Wage Dynamics, 2019-2024

The cross-sectional analysis in Section 4 establishes that AI exposure is concentrated in high-wage, cognitive occupations. A natural follow-up is whether high-exposure occupations are already experiencing differential employment or wage trends. I exploit the panel structure created by matching BLS OEWS data from May 2019 and May 2024 at the six-digit SOC level. This five-year window captures the period spanning the development and widespread deployment of large language models and generative AI tools, including ChatGPT (November 2022) and GPT-4 (March 2023).

5.1 Panel Construction

The panel matches 661 occupations observed in both the May 2019 and May 2024 OEWS releases with AIOE scores from Felten et al. (2021). The key outcome variables are the change in log employment and the change in log annual mean wages between the two periods. Occupations are split into high- and low-exposure groups at the median AIOE (−0.156), and also divided into terciles for finer-grained comparison.

5.2 First-Difference Regressions

The first-difference approach regresses Δlog(Y) between 2019 and 2024 on AIOE: $\Delta\log(Y_i) = \alpha + \beta \cdot AIOE_i + \varepsilon_i$. Under the assumption that unobserved time-invariant occupation characteristics are differenced out, this provides a quasi-causal estimate of whether higher AI exposure is associated with differential labor market trajectories.

For employment, the bivariate first-difference regression yields a positive and significant coefficient: $\beta = 0.052$ (SE = 0.009, $p < 0.001$, $R^2 = 0.046$), indicating that a one-unit increase in AIOE is associated with 5.2 log points of additional employment growth over the five-year period. However, after controlling for baseline log employment and log wages, the coefficient collapses to 0.004 and becomes statistically insignificant ($p = 0.719$), suggesting that the bivariate association is partly driven by compositional differences in occupation size and pay level at baseline.

For wages, the pattern reverses. The bivariate regression shows $\beta = -0.029$ (SE = 0.003, $p < 0.001$, $R^2 = 0.123$): higher-AIOE occupations experienced significantly slower wage growth. This result is robust to controls, with baseline employment and wages included, the coefficient attenuates modestly to −0.016 but remains significant ($p = 0.004$). Table 2 reports the first-difference regression results.

Table 2. First-Difference Regressions: 2019-2024 Changes in Log Employment and Log Wages on AIOE

	Model 4	**Model 5**	**Model 6**	**Model 7**
Dep. Variable	Δlog(Emp)	Δlog(Emp)	Δlog(Wage)	Δlog(Wage)
AIOE	0.05180***	0.00385	−0.02946***	−0.01641**
	(0.00909)	(0.01070)	(0.00334)	(0.00568)
Baseline controls	No	Yes	No	Yes
R-squared	0.046	0.144	0.123	0.196
N	661	661	661	661

Robust (HC1) standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

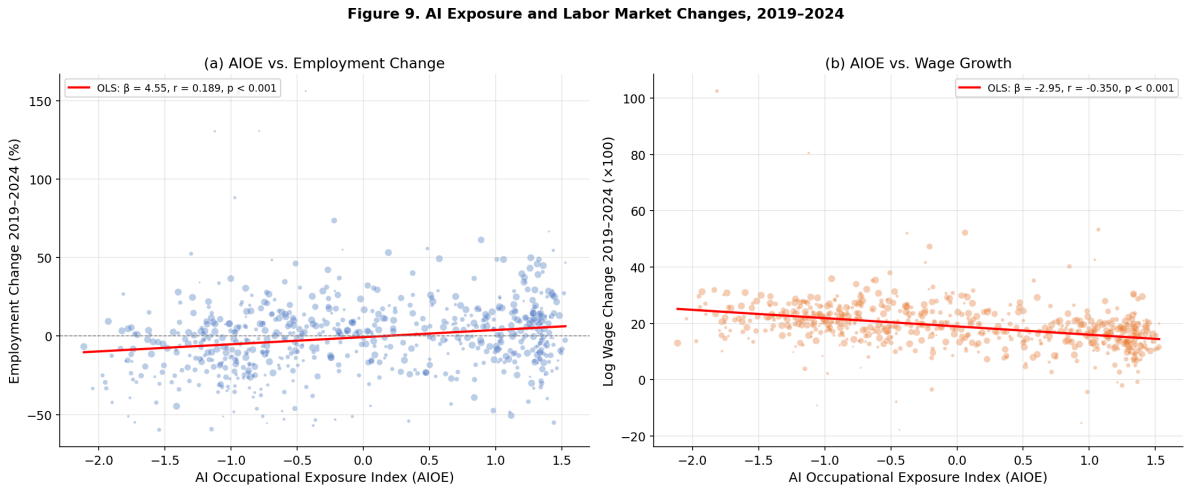

Figure 9. AI exposure vs. employment change (left) and wage growth (right), 2019-2024. Each point is an occupation; size reflects 2019 employment. Red lines are OLS fitted values.

5.3 Panel Difference-in-Differences

The formal DiD specification stacks the 2019 and 2024 observations in long format and estimates $Y_{it} = \alpha + \beta_1 \cdot AIOE_i + \beta_2 \cdot Post_t + \beta_3 \cdot (AIOE_i \times Post_t) + \varepsilon_{it}$, with standard errors clustered at the occupation level.

For log employment, the DiD interaction is positive and significant: $\beta_3 = 0.052$ (SE = 0.009, $p < 0.001$). Relative to low-exposure occupations, high-exposure occupations experienced approximately 5.2 log points faster employment growth between 2019 and 2024. The Post coefficient is −0.036 ($p < 0.001$), reflecting a modest average decline in occupation-level employment counts over the period (partly attributable to OEWS methodology changes and post-pandemic composition shifts).

For log wages, the DiD interaction is negative and significant: $\beta_3 = -0.029$ (SE = 0.003, $p < 0.001$). High-AIOE occupations experienced smaller wage increases than low-AIOE occupations, even as the overall Post coefficient shows substantial wage growth (0.189, $p < 0.001$), reflecting economy-wide nominal wage inflation. Table 3 presents the full results.

Table 3. Panel Difference-in-Differences Regressions

	Log(Employment)	**Log(Wages)**
Constant	10.78594***	10.92309***
	(0.06322)	(0.01433)
AIOE	0.07255	0.25272***
	(0.06360)	(0.01285)
Post (2024)	−0.03573***	0.18890***
	(0.00926)	(0.00307)
AIOE × Post (DiD)	**0.05180*** **	**−0.02946*** **
	(0.00909)	(0.00334)
R-squared	0.004	0.342

N	1,322	1,322

Occupation-clustered standard errors in parentheses. * p < 0.05, ** p < 0.01, *** p < 0.001.

5.4 Tercile Comparisons

The tercile analysis confirms a monotonic gradient. Mean log employment change ranges from −0.112 in the low-AIOE tercile (n = 221) to −0.013 in the medium tercile (n = 220) to +0.014 in the high-AIOE tercile (n = 220). Conversely, mean log wage change ranges from +0.221 in the low-AIOE tercile to +0.195 in the medium tercile to +0.153 in the high-AIOE tercile. Two-sample t-tests comparing the high- and low-exposure halves confirm both differences are statistically significant: for employment change, t = 4.48 (p < 0.001); for wage change, t = −7.98 (p < 0.001). Figure 10 displays these patterns with 95% confidence intervals.

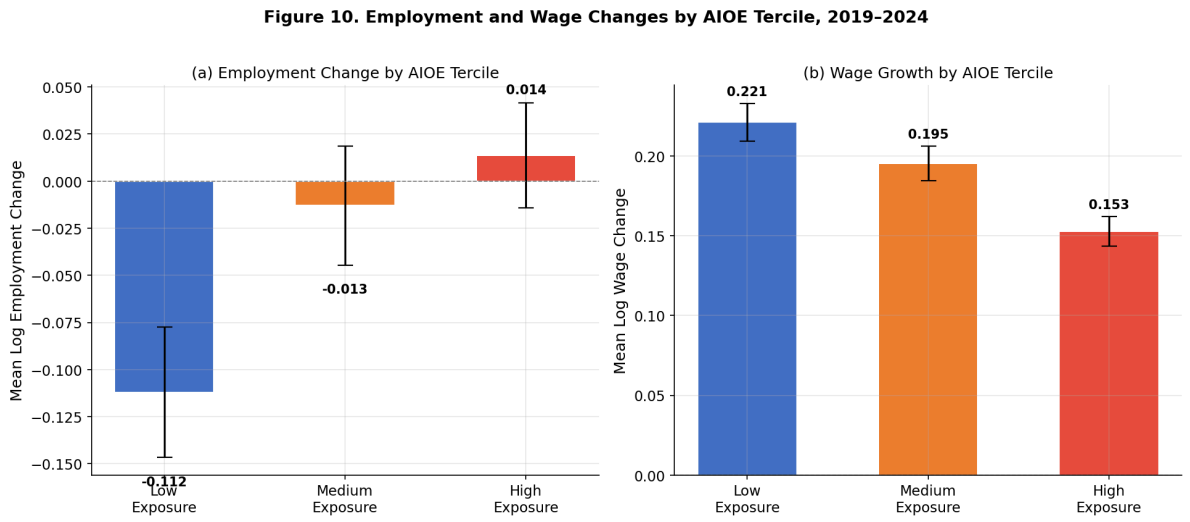

Figure 10. Employment change (left) and wage growth (right) by AIOE tercile, 2019-2024. Error bars show 95% CIs.

5.5 Winners and Losers Among High-AIOE Occupations

The aggregate DiD estimates mask considerable within-group heterogeneity. Among occupations with AIOE above 0.5, employment outcomes are strikingly bimodal. The largest declines, credit authorizers (−55%), telemarketers (−51%), switchboard operators (−48%), statistical assistants (−40%), and proofreaders (−33%), are all routine cognitive roles whose tasks AI can directly substitute. The largest gains, industrial-organizational psychologists (+67%), epidemiologists (+55%), sales managers (+50%), computer and information systems managers (+49%), and marketing managers (+46%), are roles in which AI likely complements rather than replaces human expertise. This bifurcation underscores that exposure alone does not mechanically translate to displacement; the crucial distinction is whether AI substitutes for or augments the specific task bundle of each occupation.

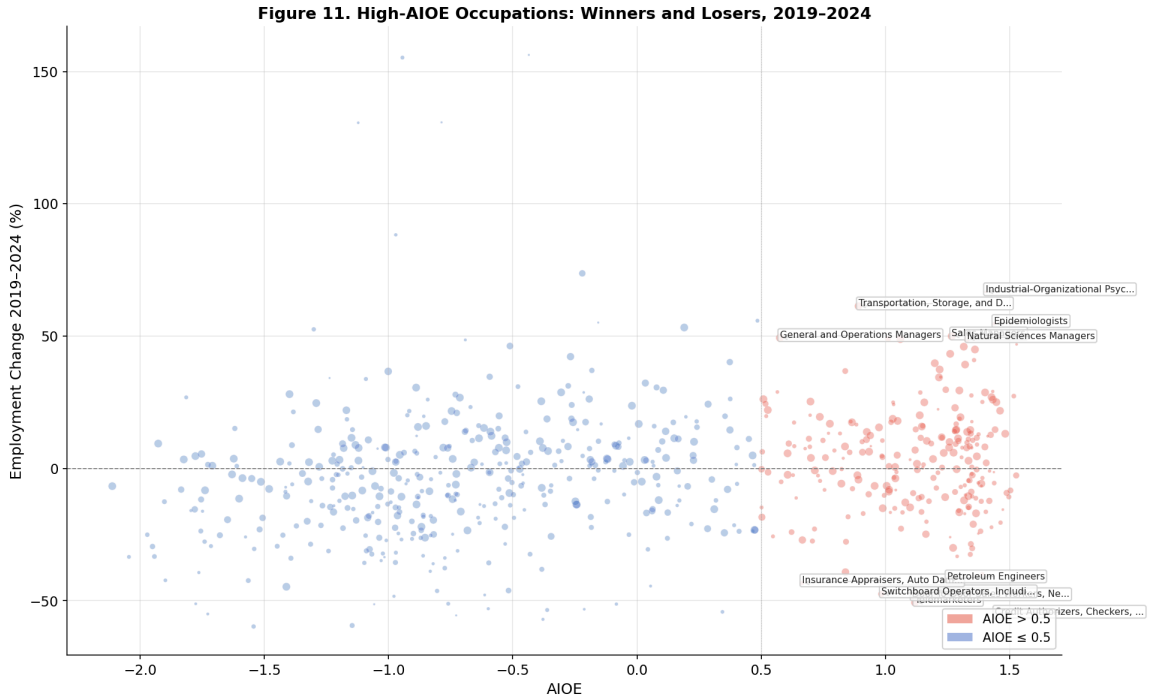

Figure 11. High-AIOE occupations show heterogeneous employment outcomes. Red points denote AIOE > 0.5. Annotations highlight the largest gainers and decliners.

5.6 Interpretation of DiD Results

The DiD results paint a nuanced picture. High-AIOE occupations are not, on average, contracting, in fact, they are growing faster than low-AIOE occupations. However, their wage growth has been meaningfully slower, consistent with theories in which AI augments cognitive labor supply, raises effective productivity of exposed workers, and attracts new entrants, thereby moderating wage premiums. The compression of the AI skill premium, even as employment expands, may reflect the early stages of a structural shift in which AI tools raise cognitive-work productivity, expand output capacity, and dampen wage growth relative to less-exposed sectors experiencing labor shortages (e.g., construction, transportation).

These findings are broadly consistent with Kolko's (2026) assessment that current evidence is mixed and early. The employment-growth result aligns with the complementarity hypothesis emphasized by Brynjolfsson and Mitchell (2017), while the wage-compression result raises concerns by Acemoglu and Restrepo (2020) about the distributional consequences of automation. The heterogeneity within high-AIOE occupations, routine cognitive roles decline while managerial and analytical roles grow, supports task-level analyses that distinguish between substitution and augmentation effects and cautions against interpreting any single aggregate coefficient as the full story.

6. Discussion

6.1 Mechanisms and Interpretation

The central substantive finding is that AI exposure follows a fundamentally different occupational gradient than previous automation technologies. Where industrial robots primarily displaced middle-skill

routine workers (Acemoglu and Restrepo, 2020), AI capability overlaps most with high-wage, cognitively intensive occupations. However, high exposure does not necessarily imply displacement. The combination of faster employment growth and slower wage growth in high-AIOE occupations is most parsimoniously rationalized by an augmentation mechanism: AI lowers the per-unit cost of producing cognitive output, expands the supply of effective knowledge work, and compresses the premium paid for such work even as aggregate demand for it continues to grow. This is the reverse of what a pure displacement story would predict, in which we would expect simultaneous employment declines and wage declines.

6.2 Distributional Stakes

The gender composition result deserves greater prominence in policy discussions. The AIOE gap between female and male dominated occupation groups is more than 1.2 standard deviations of the exposure distribution. If the wage-compression effect documented here persists or intensifies, AI-driven structural change could disproportionately attenuate wage gains in occupations where women have built a durable labor market presence since the 1980s, with implications for the gender pay gap that go beyond the usual story of glass ceilings and caregiving penalties.

6.3 Limitations

These results should be read as descriptive quasi-experimental rather than as clean causal identification. Four limitations deserve emphasis. First, the AIOE measures exposure based on AI capabilities as of 2021 and does not fully capture generative-AI advances since. Second, the two-period DiD design cannot test the parallel-trends assumption directly; occupation-specific pre-trends, for instance, routine-biased shifts already underway before 2019, could confound the interpretation. Third, the demographic analysis relies on occupation-level gender composition rather than individual-level microdata, limiting subgroup inference. Fourth, the outcome variable is occupation-level employment counts, not worker-level displacement flows; behind every constant occupation count is turnover that these data cannot observe.

6.4 Future Work

Extensions include incorporating generative-AI-specific exposure scores (e.g., Eloundou et al. 2023), extending the panel to additional OEWS waves (2020, 2021, 2022, 2023) to test pre-trends and implement event-study specifications around the November 2022 ChatGPT release, integrating CPS microdata for individual-level demographic analysis, and replicating the design on Canadian NOC occupations to test generalizability to a different labor market with different AI adoption timing.

7. Conclusion

This paper provides a comprehensive descriptive and quasi-experimental analysis of AI occupational exposure across 673 U.S. occupations. Three findings stand out. First, AI exposure is concentrated in high-wage, cognitively intensive occupations, a fundamental departure from the routine-biased gradient of earlier automation. Second, exposure is distributionally skewed, with female-dominated occupations facing substantially higher AI exposure than male-dominated ones. Third, and most importantly, the difference-in-differences analysis of 661 occupations tracked from 2019 to 2024 reveals faster employment growth but slower wage growth in high-exposure occupations, a pattern most consistent with

augmentation-plus-supply-expansion rather than wholesale displacement. The heterogeneity within high-AIOE occupations, with routine cognitive roles contracting while managerial and research roles expand, underscores that task-level substitution-versus-complementarity dynamics, not aggregate exposure alone, will determine who benefits from and who is displaced by the next phase of AI deployment.

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Generative AI Disclosure

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